

Graph Neural Networks for Fraud Detection: A Survey and an Imbalance-Aware Approach

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Abstract—Fraud detection in online services can be formulated as semi-supervised node classification on graphs, where fraudsters are heavily outnumbered and actively camouflage themselves among benign users. This report surveys representative graph-based fraud detection methods — from the dense-block detector FRAUDAR to graph neural network (GNN) approaches such as SemiGNN, CARE-GNN, and PC-GNN — and proposes an imbalance-aware GNN that combines label-supervised similarity filtering of camouflaged neighbors with balanced neighborhood sampling. Experiments on the YelpChi and Amazon fraud benchmarks show that the proposed model outperforms a GCN baseline by a large margin and surpasses SemiGNN on YelpChi across all metrics (0.854 vs. 0.801 AUROC, mean over three seeds), while remaining competitive on Amazon. Ablation and interaction studies show that the similarity filter drives the improvement, whereas balanced sampling adds no measurable gain once class-weighted training is used. Code and results are publicly available.

Index Terms—graph neural networks, fraud detection, anomaly detection, class imbalance, semi-supervised learning

I. INTRODUCTION

Online review and e-commerce platforms lose revenue and user trust to fraudulent accounts. Because fraudsters deliberately mimic benign behavior (*camouflage*) [1], purely feature-based detectors are easily evaded. Modeling users, reviews, and products as a graph and applying graph neural networks (GNNs) has therefore become a dominant paradigm; deep graph anomaly detection more broadly is surveyed in [2]. Two challenges are central: (1) extreme class imbalance between fraudsters and benign users, and (2) camouflage along both feature and relation dimensions. This project studies GNN-based fraud detectors that explicitly address these challenges.

Concretely, this report makes three contributions. First, it surveys representative graph-based fraud detection methods from top venues and organizes them along the two challenges above (Section II). Second, it proposes a simple imbalance-aware GNN that replaces CARE-GNN’s reinforcement-learning neighbor selector with a fixed, label-supervised similarity filter, and combines it with PC-GNN-style balanced sampling (Section IV). Third, it evaluates the model on two public benchmarks with three random seeds and reports ablation and interaction studies that isolate the contribution of each component (Section V).

II. RELATED WORK

A. Graph-based Fraud Detection without Deep Learning

A long line of work detects fraud through purely structural signals: groups of fraudulent accounts act in lockstep, creating unusually dense blocks in the adjacency matrix of, e.g., a user-product review graph. Density-based detectors exploit exactly this signal, but fraudsters evade them through *camouflage*: adding edges to honest, popular targets so as to “look normal.” FRAUDAR [1] is the representative method of this family. It detects dense subgraphs with a greedy, near-linear-time algorithm whose suspiciousness metric is designed to be robust to camouflage edges, and it provides provable, data-dependent bounds on the amount of fraud that can remain undetected on a given graph. In experiments, FRAUDAR detected injected fraud on a Twitter graph with high accuracy even under camouflage, where earlier density-based methods fail. Such methods are scalable and effective, but they ignore node features and the (scarce) available labels, which limits their accuracy on attributed graphs.

B. Deep Graph Anomaly Detection

Ma et al. [2] systematize deep learning approaches to graph anomaly detection, organizing methods by the anomalous object they target (node, edge, subgraph, or graph level) and by architecture. Their survey highlights two challenges that are central to fraud detection and to this work: (1) extreme *class imbalance*, since anomalies are by definition rare, and (2) *camouflage*, since adversarial anomalies deliberately resemble normal instances. Fraud detection on review or transaction networks corresponds to node-level anomaly detection on attributed, multi-relation graphs under semi-supervision — the setting adopted by all methods below.

C. GNN-based Fraud Detectors

Graph neural networks (GNNs) detect fraudulent nodes by aggregating neighborhood information. SemiGNN [3] is a semi-supervised graph attentive network developed for financial fraud detection at payment-platform scale: it organizes each node’s neighborhood into multiple views, applies hierarchical attention (node-level attention within a view, then view-level attention across views), and exploits unlabeled nodes through a graph-regularized semi-supervised loss that pulls

TABLE I
SURVEYED METHODS AND THE CHALLENGES THEY ADDRESS.

Method	Venue	Camouflage	Imbalance
FRAUDAR [1]	KDD'16	yes	no
SemiGNN [3]	ICDM'19	no	partial
CARE-GNN [4]	CIKM'20	yes	no
PC-GNN [5]	WWW'21	no	yes
Ours	—	yes	yes

together embeddings of adjacent nodes. This semi-supervision is essential in practice, since labeled fraudsters are scarce and expensive to obtain.

CARE-GNN [4] directly targets camouflage, distinguishing *feature camouflage* (fraudsters craft feature vectors similar to benign users) and *relation camouflage* (fraudsters link to benign entities). It introduces a label-aware similarity measure to identify informative neighbors, uses reinforcement learning (RL) to adapt a per-relation filtering threshold, and aggregates the filtered neighborhoods relation-wise. CARE-GNN also released the YelpChi and Amazon fraud benchmarks used in this study, on which it clearly outperforms vanilla GCN and GAT. Its main drawbacks are the cost and instability of the RL-based selector and that class imbalance is handled only implicitly.

D. Imbalance-aware Graph Learning for Fraud Detection

Fraudsters typically constitute well under 15% of nodes, so naive GNN training is dominated by the majority class. PC-GNN [5] addresses this with a “pick and choose” strategy: a label-balanced sampler *picks* a balanced subset of labeled nodes for each training step, and a neighborhood sampler *chooses* neighbors by thresholding feature-space distances per relation, so that aggregation uses a balanced and informative neighborhood. It reports consistent gains on the YelpChi and Amazon benchmarks under skewed label distributions. A complementary, architecture-agnostic line of work handles imbalance at the loss level with class-weighted or focal objectives; we adopt a class-weighted cross-entropy in this work and study how it interacts with sampling-level corrections.

E. Summary and Positioning

Table I contrasts the surveyed methods along the two challenges identified above. No single method addresses both camouflage and imbalance in a simple, stable way: FRAUDAR handles camouflage but ignores features and labels; CARE-GNN handles camouflage with an expensive RL selector; PC-GNN handles imbalance at the sampling level but ignores camouflage. Our approach (Section IV) combines a fixed, label-supervised similarity filter in the spirit of CARE-GNN — removing the RL machinery — with PC-GNN-style balanced neighborhood sampling and a class-weighted objective, and we ablate each component to measure its contribution.

III. PROBLEM DEFINITION

Let $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ be an attributed graph with node set $\mathcal{V}$, edge set $\mathcal{E}$, and node feature matrix $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$. In

our benchmarks the graph is multi-relation: $\mathcal{E}$ is the union of relation-specific edge sets $\mathcal{E}_1, \dots, \mathcal{E}_R$ ($R = 3$). Each node $v \in \mathcal{V}$ has a label $y_v \in \{0, 1\}$ ($1 = \text{fraudster}$), of which only a small labeled subset $\mathcal{V}_L \subset \mathcal{V}$ is observed, with $|\{v : y_v = 1\}| \ll |\{v : y_v = 0\}|$ (class imbalance). The task is *semi-supervised node classification*: learn $f : \mathcal{V} \rightarrow [0, 1]$ from G and $\mathcal{V}_L$ that ranks fraudulent nodes above benign ones on the unlabeled nodes.

IV. PROPOSED METHOD

Our model, an *imbalance-aware multi-relation GNN*, has three components: (A) camouflage-resistant neighbor filtering, (B) balanced neighborhood sampling, and (C) relation-wise attention aggregation and fusion. We describe each component formally.

A. Camouflage-Resistant Neighbor Filtering

Following CARE-GNN [4], we assume camouflaged neighbors are feature-wise dissimilar to the center node. For each relation r , a two-layer MLP ϕ_r scores the similarity between the center node u and a neighbor v from their projected features $\mathbf{h}_u, \mathbf{h}_v$:

$$s_r(u, v) = \sigma(\phi_r([\mathbf{h}_u \| \mathbf{h}_v])). \quad (1)$$

Instead of CARE-GNN’s RL controller, we keep a fixed fraction $\rho = 0.5$ of each node’s neighbors per relation (the top-scored ones), which removes the RL training loop entirely and is stable in practice. Because hard top- k selection is non-differentiable, ϕ_r receives no gradient from the classification loss; as in CARE-GNN, we therefore train it with label supervision: a binary cross-entropy loss $\mathcal{L}_{\text{sim}}$ that predicts, for edges incident to labeled training nodes, whether the two endpoints share the same label.

B. Balanced Neighborhood Sampling

Following PC-GNN [5], we rebalance the aggregated neighborhood toward the minority class. Each node’s filtered neighborhood is capped at $K = 32$ neighbors, sampled without replacement with probability proportional to a weight w_v : $w_v = \beta = 4$ for neighbors known (from *training* labels) to be fraudulent, and $w_v = 1$ otherwise. Sampling uses Efrimidis–Spirakis keys $\log u/w_v$, $u \sim \mathcal{U}(0, 1)$, which gives exact weighted sampling without replacement; at evaluation time sampling is deterministic (key $\log w_v$).

C. Relation-wise Aggregation and Fusion

Within each relation, the filtered and sampled neighborhood is aggregated with dot-product attention:

$$\alpha_{uv} = \text{softmax}_v \left(\frac{\mathbf{h}_u^\top \mathbf{h}_v}{\sqrt{d}} \right), \quad \mathbf{h}_u^{(r)} = \sum_{v \in \hat{\mathcal{N}}_r(u)} \alpha_{uv} \mathbf{W} \mathbf{h}_v, \quad (2)$$

where $\hat{\mathcal{N}}_r(u)$ is the filtered, sampled neighborhood. A residual connection preserves the center node’s own information, $\tilde{\mathbf{h}}_u^{(r)} = \mathbf{h}_u^{(r)} + \mathbf{h}_u$, and a learned cross-relation attention vector fuses the R relation embeddings into the final representation, which a linear layer maps to two logits.

TABLE II
DATASET STATISTICS (FRAUD RATE IS OVER LABELED NODES).

	YelpChi	Amazon
Nodes	45,954	11,944
Edges (all relations)	7,693,758	8,796,784
Features	32	25
Fraud rate	14.5%	9.5%

D. Training Objective

The model is trained with class-weighted cross-entropy $\mathcal{L}_{CE}^w$ (inverse class frequency, computed on the training split) plus the similarity-supervision term:

$$\mathcal{L} = \mathcal{L}_{CE}^w + \lambda \mathcal{L}_{sim}, \quad \lambda = 0.1. \quad (3)$$

All components are trained end-to-end with Adam.

V. EXPERIMENTS

A. Datasets

We use two public multi-relation fraud benchmarks released with CARE-GNN [4] (Table II). **YelpChi** contains hotel and restaurant reviews labeled as spam or genuine, with relations R-U-R (same user), R-S-R (same star rating), and R-T-R (same product and time window). **Amazon** contains product reviewers labeled as fraudsters or benign, with relations U-P-U (co-reviewed product), U-S-U (same star rating), and U-V-V (similar review text). Following CARE-GNN, features are L2-normalized, splits are stratified with 40% train, and Amazon’s unlabeled nodes are excluded from the splits but kept in the graph for message passing; we carve the remaining 60% into 20% validation and 40% test.

B. Experimental Setup

We compare our model against two baselines: a two-layer **GCN** on the union graph, and **SemiGNN** [3] (implemented per the paper, with its semi-supervised auxiliary loss). All models use hidden dimension 64, Adam with learning rate 0.01, 100 epochs, and class-weighted cross-entropy; the best checkpoint by validation AUROC is evaluated on the test split. We report AUROC, AUPRC, F1-macro, and Recall@100, the standard metrics for this benchmark [4], [5], as mean $\pm$ standard deviation over three seeds (1, 7, 42). All experiments run on CPU; a full training run takes 15–60 minutes depending on the model.

C. Main Results

Table III shows the main results. Both relation-aware models vastly outperform GCN on YelpChi (GCN remains close to random at 0.585 AUROC), confirming that flat homophily-based aggregation fails under camouflage. On YelpChi our model is the best on every metric: it improves over SemiGNN by +5.3 AUROC and +15.2 F1-macro, a gap far larger than the seed-to-seed standard deviation (≤ 0.01). On Amazon, SemiGNN keeps a narrow edge in ranking metrics (AUROC/AUPRC), while our model achieves the best F1-macro and Recall@100. We attribute the difference to the structure of the Amazon graph, where a single relation (U-S-U) carries

TABLE III
TEST RESULTS (MEAN $\pm$ STD OVER 3 SEEDS). BEST PER COLUMN IN BOLD.

Dataset	Model	AUROC	AUPRC	F1-macro
YelpChi	GCN	0.585 $\pm$.002	0.216 $\pm$.008	0.443 $\pm$.088
	SemiGNN	0.801 $\pm$.008	0.407 $\pm$.012	0.540 $\pm$.000
	Ours	0.854$\pm$.003	0.518$\pm$.006	0.692$\pm$.003
Amazon	GCN	0.855 $\pm$.002	0.444 $\pm$.008	0.621 $\pm$.012
	SemiGNN	0.959$\pm$.004	0.879$\pm$.009	0.878 $\pm$.011
	Ours	0.944 $\pm$.002	0.858 $\pm$.004	0.904$\pm$.003

TABLE IV
ABLATION OF OUR MODEL (SEED 42).

Dataset	Variant	AUROC	AUPRC	F1-macro
YelpChi	Ours (full)	0.856	0.521	0.694
	Ours $_{\neg}$ filter	0.811	0.449	0.613
	Ours $_{\neg}$ sampler	0.855	0.519	0.699
Amazon	Ours (full)	0.943	0.853	0.902
	Ours $_{\neg}$ filter	0.948	0.856	0.779
	Ours $_{\neg}$ sampler	0.943	0.853	0.902

over 80% of edges, leaving less room for similarity filtering to help.

D. Ablation Study

Table IV ablates the two components of our model. Removing the similarity filter (Ours $_{\neg}$ filter, which also drops $\sim 25K$ parameters) degrades YelpChi AUROC by 4.5 points and F1-macro by 8.1 points, showing that camouflage-resistant filtering is the main driver of the improvement. On Amazon, removing the filter slightly raises AUROC but cuts F1-macro by 12 points: filtering mainly improves *decision* quality rather than *ranking* quality there. Removing balanced sampling (Ours $_{\neg}$ sampler) changes results by less than 0.1 AUROC points on both datasets — the component contributes nothing in this setting, a negative result we analyze next.

E. Interaction of Imbalance Mechanisms

To test whether balanced sampling is merely redundant with class-weighted training, we ran a 2×2 study crossing the two imbalance mechanisms (sampling vs. class-weighted loss). With class weights disabled, the sampler still changes results by ≤ 0.2 AUROC points on both datasets, while removing class weights on YelpChi collapses F1-macro from 0.69 to 0.46. We conclude that (i) loss-level reweighting is the effective imbalance mechanism on these benchmarks, and (ii) PC-GNN-style sampling does not transfer to this setting, plausibly because known-fraud neighbors are too rare (9–15% base rate) for a $4\times$ sampling boost to alter aggregated neighborhoods materially. We view reporting such negative results as important for reproducible fraud-detection research.

VI. CONCLUSION AND FUTURE WORK

We surveyed representative graph-based fraud detection methods and proposed a simple imbalance-aware GNN that replaces CARE-GNN’s RL neighbor selector with a fixed,

label-supervised similarity filter. The model is the best performer on YelpChi across all metrics and achieves the best F1-macro on Amazon, and our studies identify the similarity filter as the decisive component. Future work includes heterophily-aware aggregation, tuning or removing the balanced sampler (e.g., larger boost factors or learnable sampling), extending the ablations to multiple seeds, and evaluating on additional datasets such as Elliptic.

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